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The technology acceptance model (TAM): A meta-analytic structural equation modeling approach to explaining teachers' adoption of digital technology in education

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Highlights

- The Technology Acceptance Model (TAM) explains teachers' technology adoption.
- Relations among variables in the TAM are synthesized meta-analytically.
- The TAM and its versions fit the data well—even for subsamples of teachers.

- Within the TAM, direct effects of PU on BI and ATT on USE exist.
- The TAM explains behavioral intentions and technology use significantly.

Abstract

The extent to which teachers adopt technology in their teaching practice has long been in the focus of research. Indeed, a plethora of models exist explaining influential factors and mechanisms of technology use in classrooms, one of which—the Technology Acceptance Model (TAM) and versions thereof—has dominated the field. Although consensus exists about which factors in the TAM might predict teachers' technology adoption, the current field abounds in some controversies and inconsistent findings. This meta-analysis seeks to clarify some of these issues by combining meta-analysis with structural equation modeling approaches. Specifically, we synthesized 124 correlation matrices from 114 empirical TAM studies (N = 34,357 teachers) and tested the fit of the TAM and its versions. Overall, the TAM explains technology acceptance well; yet, the role of certain key constructs and the importance of external variables contrast some existing beliefs about the TAM. Implications for research and practice are discussed.

Introduction

Technology pervades almost all areas in society. Considering education, at least two trends can be observed: First, educational systems around the world are incorporating digital competences in curricula and assessments (Beller, 2013; Flórez et al., 2017; Siddiq, Hatlevik, Olsen, Throndsen, & Scherer, 2016). Second, teachers and teacher educators are encouraged to include technology in their teaching—as a tool to facilitate learning or as a means to formative assessment (Shute & Rahimi, 2017; Straub, 2009). It has become the designated aim of education to help students to become digitally literate citizens who can cope with the complexities and dynamics in today's societies (Fraillon, Ainley, Schulz, Friedman, & Gebhardt, 2014). This development, however, necessitates the meaningful inclusion of technology in teaching and learning contexts (OECD, 2015; Siddiq, Scherer, & Tondeur, 2016). An extensive body of literature has dealt with the factors associated with this inclusion by focusing on teachers' adoption of technology (Straub, 2009). One model

though has dominated the research landscape—the Technology Acceptance Model (TAM). The TAM comprises several variables explaining behavioral intentions and the use of technology directly or indirectly (i.e., perceived usefulness, perceived ease of use, attitudes toward technology), and has been extended by external variables, such as self-efficacy, subjective norms, and facilitating conditions of technology use (Schepers & Wetzels, 2007). The TAM has gained considerable prominence, particularly due to its transferability to various contexts and samples, its potential to explain variance in the intention to use or the use of technology, and its simplicity of specification within structural equation modeling frameworks (e.g., King & He, 2006; Marangunić & Granić, 2015). Besides, the TAM is a powerful vehicle to describe teachers' technology adoption next to other models.

Despite its prominence, however, the existing body of research does not draw a clear picture about specific relations within the TAM: Whereas some studies confirmed the hypothesized relations fully, others did not (King & He, 2006; Šumak, Heričko, & Pušnik, 2011). This finding is further substantiated by significant variation of TAM relations across studies and samples, and consequently calls for a systematic synthesis. Furthermore, whereas previous meta-analyses on the TAM included a large variety of samples from multiple occupations and domains (Hsiao & Yang, 2011; Ritter, 2017; Schepers & Wetzels, 2007), a systematic review of the TAM for teachers in educational contexts is, to our best knowledge, lacking. It is important to synthesize the existing findings on teachers' technology acceptance though, because they provide further insights into the possible mechanisms behind technology acceptance—insights relevant to teacher education and professional development. The current meta-analysis consequently reviews studies presenting the TAM for teacher samples. We take a meta-analytic structural equation modeling (MASEM) approach to synthesizing entire correlation matrices instead of single correlations and further quantify their variation across teacher samples, particularly for pre- and in-service teachers. Besides, we explore model fit, moderation effects, and the effects of external variables within the TAM.

Education has always lived in tension between two functions: education as a matter of assuring continuity and as a matter of fostering creativity and change. Within these, technology brings a new set of challenges and pressures for educational institutions (Romeo, Lloyd, & Downes, 2013). The speed with which the evolution of technology has taken place is phenomenal. Today, school teachers in many countries around the world are working with “digital natives” who are growing up with new technologies as a non-remarkable feature of their life. Technology allows us to (co-)create, collect, store and use

knowledge and information; it enables us to connect with people and resources all over the world, to collaborate in the creation of knowledge and to distribute and benefit from knowledge products (Spector, 2008; von Davier, Hao, Liu, & Kyllonen, 2017).

The question remains as to what degree teachers integrate technology into teaching and learning activities. Research reveals that integrating technology is a complex process of educational change, and the extent of technological applications in schools is still extremely varied (Bishop & Spector, 2014; Fraillon et al., 2014). Clearly, emerging educational technology usage in (teacher) education has increased in recent years, but technology acceptance and usage continue to be problematic for educational institutions (Berrett, Murphy, & Sullivan, 2012; Straub, 2009). In the literature, the question is repeatedly put forward as to what variables determine technology integration in education. Measuring user acceptance of technology is a way of determining the teacher's intentions toward using new technologies in their educational practice. Over the last decades, a series of models have been proposed to describe the mechanism behind and factors affecting technology adoption, such as the Unified Theory of Acceptance and Use of Technology (UTAUT) and the Technology Acceptance Model. These models have emerged from well-established psychological theories, including the Theory of Reasoned Action (Fishbein, 1979) and the Theory of Planned Behavior (Ajzen, 1991). The UTAUT, for example, describes four core determinants of the user intentions and the actual use of technology, namely performance and effort expectancy, social influence, and facilitating conditions (Venkatesh, Morris, Davis, & Davis, 2003). The effects of these determinants are hypothesized to be moderated by respondents' gender, age, experience, and the voluntariness of technology use (Williams, Rana, & Dwivedi, 2015). The setup of the UTAUT is comparable to that of the TAM, and the determinants share similarities in their conceptualization (Nistor & Heymann, 2010). Although it is more difficult to test than the TAM (due to the hypothesized moderation effects), this model is considered another, powerful model describing technology acceptance. The TAM and the UTAUT are only two examples of technology acceptance models; several extensions and alternatives have evolved over time (for a comprehensive overview, please review Taherdoost, 2018). Despite the variety of models, the TAM has dominated the research landscape as the most commonly used model to describe use intentions and actual technology use (Hsiao & Yang, 2011; King & He, 2006; Marangunić & Granić, 2015).

At the same time, the TAM falls short of conceptualizing what it means to accept and integrate technology in classrooms. More specifically, the model does not specify which types of professional knowledge about teaching and learning with technology teachers

must have in order to integrate technology meaningfully. These types of knowledge are specified in the so-called Technological Pedagogical Content Knowledge (TPACK) framework, a framework that defines different kinds of knowledge domains teachers need to become proficient in for successfully integrating digital technology in teaching and learning processes (Koehler, Mishra, Kereluik, Shin, & Graham, 2014). These knowledge domains comprise content knowledge, pedagogical knowledge, and pedagogical content knowledge in the context of technology, as well as the mere technological knowledge (Mishra & Koehler, 2006). Koehler and Mishra (2009) argued that, for technology integration to occur in education, teachers must be competent in these forms of knowledge, but more importantly, they must be able to integrate all types of knowledge. In other words, TPACK emphasizes the importance of preparing pre-service teachers to make sensible choices in their uses of technology when teaching particular content to a specific target group, as it can lead to a better understanding about how teachers make decisions that affect technology acceptance and integration into teaching and learning processes. From this perspective, it is anticipated that teachers will be likely to "accept" a new technology once they perceive it as relevant for specific didactical approaches within their subjects. In addition, Mei, Brown, and Teo (2017) found in their study that teachers who perceived themselves as competent in the TPACK domains were more likely to accept and integrate technology in their teaching. Hsu (2016) further found that both PEU and PU can be predicted by TPACK. Considering this, a link to the TPACK framework could address the shortcoming of the TAM and enhance the understanding of technology acceptance processes.

The Technology Acceptance Model, first proposed by Davis (1985), comprises core variables of user motivation (i.e., perceived ease of use, perceived usefulness, and attitudes toward technology) and outcome variables (i.e., behavioral intentions, technology use). Of these variables, perceived usefulness (PU) and perceived ease of use (PEU) are considered key variables that directly or indirectly explain the outcomes (Marangunić & Granić, 2015). These variables are often accompanied by external variables explaining variation in perceived usefulness and ease of use: Among others, subjective norms (SN), self-efficacy (CSE), and facilitating conditions (FC) were significantly related to the TAM core variables—however, to different degrees (Abdullah & Ward, 2016; Schepers & Wetzels, 2007). These external variables represent personal capabilities next to contextual factors. Their conceptualizations, however, vary across studies and thus necessitate clear definitions in the current meta-analysis. We present the definitions applied to this meta-analysis in Table 1. Overall, perceived ease of use and perceived usefulness, the most important factors in the TAM, refer to the degrees to

which a person believes that using technology would be free from effort (PEU) and that using technology would enhance their job or task performance (PU). In this context, “free from effort” means “free from difficulty or great effort”, as Davis (1989) in his seminal paper specified. PEU therefore refers to the effort a person estimates it would take to use technology and is closely related to competence beliefs (Scherer, Siddiq, & Teo, 2015). These two perceptions, PEU and PU, directly relate to another TAM-core variable, attitudes toward technology (ATT). Most commonly, the TAM comprises at least one outcome variable: behavioral intention (BI) and/or technology use (USE). Inspired by the Theory of Reasoned Action, the former refers to intended behavior, whereas the latter refers to observable behavior, that is, the actual use of technology. In most versions of the TAM, BI predicts USE—however, the direction of this link is not deterministic because positive user experience may also determine future behavioral intentions (Straub, 2009). Finally, external variables in the TAM refer to perceptions of how important others consider the use of technology (SN), perceptions of one's own capabilities of mastering computer- or technology-related tasks (CSE), and perceptions of external control, that is, the organizational support for technology use (FC) in terms of organizational resources and support structures (Taylor & Todd, 1995).

Given the variety of variables within the TAM, different versions of the model have been studied empirically (Taylor & Todd, 1995). The most prominent versions are depicted in Fig. 1. Model 1 represents the TAM core and focuses on behavioral intentions as the outcome. Model 2 extends this model by technology use. Nistor (2014) noted that the link between use intentions and actual use is oftentimes missing in empirical studies of the TAM—hence, the extension of Model 1. Models 3 and 4 further add the proposed external variables to Models 1 and 2 as predictors of perceived usefulness and ease of use. This selection of TAM versions represents the typically specified path models exhibiting the hypothesized relations (Marangunić & Granić, 2015; Ritter, 2017).

Empirical research on the TAM identified several issues: First, substantial variation in specific paths in the TAM exists (Imtiaz & Maarop, 2014; T. Teo & Paul van Schaik, 2012). For instance, whereas some authors found significant direct relations between perceived usefulness and behavioral intention (e.g., E. Y. M. Cheung & Sachs, 2006; Pynoo et al., 2012), others did not (e.g., Kirmizi, 2014; Teo & Milutinovic, 2015). Second, the role of external variables explaining variation in the TAM core constructs differs (Burton-Jones & Hubona, 2006). For instance, whereas teachers’ computer self-efficacy explains considerable variation in perceived usefulness and perceived ease of use, facilitating conditions for technology use at school weakly predict these two variables—these

relations vary across studies (e.g., Nam, Bahn, & Lee, 2013; Teo & van Schaik, 2012). Third, a variety of TAM models exist, with or without external variables, with or without direct effects of certain variables on outcome variables, with or without variables grouping the teacher samples. To illustrate, Marangunić and Granić (2015) systematically reviewed research on the TAM conducted between 1986 and 2013. They identified at least three different versions of the model, some considering only USE as an outcome variable, others considering BI and USE as outcomes yet excluding ATT. Abdullah and Ward (2016) meta-analyzed a TAM version that contained external variables—the selection of variables, however, differed from that of other meta-analyses (e.g., Schepers & Wetzels, 2007). Fourth, some studies investigated the measurement and structural invariance of the TAM across groups of teachers, including pre- and in-service teachers and different nationalities (Teo, Lee, Chai, & Wong, 2009). Such studies could oftentimes not identify full invariance across groups of teachers, and the resultant findings highlight that the TAM may not fully apply to all contexts and groups of teachers to the same extent. Fifth, variables characterizing persons, contexts, and the measurement of variables may moderate the relations within the TAM (Straub, 2009).

The prominence of the TAM and the availability of primary research studies resulted in several meta-analyses that synthesized the relations and paths within the TAM in various contexts. Table 2 provides a brief account of these meta-analyses. These meta-analyses mainly focused on the TAM core variables (i.e., PEU, PU, and ATT) and outcome variables, such as behavioral intentions and technology use (Marangunić & Granić, 2015). The contexts in which the relations among them were studied vary substantially: Whereas some meta-analyses included any TAM study that had been conducted until the date of review (e.g., King & He, 2006; Schepers & Wetzels, 2007), others included only TAM studies targeted at specific educational contexts, such as e-learning platforms or instruction (e.g., Ritter, 2017; Šumak et al., 2011).

Most meta-analyses described above performed separate meta-analyses and aggregated the resultant correlations between the TAM variables in an overall correlation matrix, thus taking a univariate approach. In addition, one meta-analysis aggregated path coefficients, and one meta-analysis synthesized correlation matrices, however with a very small number of studies ($k = 13$). Although these meta-analyses provided valuable insights into the roles of certain variables in the TAM, possible group differences, and the overall variance explanation of technology use or its intentions, more recent developments of meta-analytic structural equation model (MASEM) may take these findings even further by addressing some of the challenges associated with the univariate

approaches (M. W.-L. Cheung, 2015; M. W.-L. Cheung & Chan, 2005). More specifically, the potential of MASEM procedures that combine entire correlation matrices rather than single correlations through separate meta-analyses across studies lies in the provision of more accurate correlation matrices that are further subjected to structural equation modeling. Tang and Cheung (2016), for example, showcased this benefit in the context of testing theories in internal business and warned against using univariate meta-analyses may lead to inaccurate findings.

Once correlations are pooled in previous meta-analyses, the resultant correlations, path coefficients, or correlation matrices are then submitted to moderator analyses—moderating variables target, for instance, types of users, technologies, and cultures. By and large, the effects identified in these meta-analyses suggest: (a) strong relations between PEU and PU; (b) larger effects of PU on BI than of PEU on BI; and (c) mediocre to strong ATT–BI and BI–USE relations. These effects, however, varied considerably across meta-analyses, sometimes ranging from insignificant and close-to-zero effects to strong, positive, and significant effects. This variation, in fact, points to some inconsistencies across meta-analyses, as the following example illustrates: The effects of both perceived ease of use and usefulness on teachers' attitudes toward technology differ considerably. Whereas Ritter (2017) reports a strong positive effect of PEU on ATT ($\beta_{PEU\text{-}ATT} = 0.52$) and a weak positive effect of PU on ATT ($\beta_{PU\text{-}ATT} = 0.16$), Schepers and Wetzels (2007) found the opposite ($\beta_{PEU\text{-}ATT} = 0.26$, $\beta_{PU\text{-}ATT} = 0.46$)—so did L. Zhang, Zhu, and Liu (2012) in their meta-analysis ($\beta_{PEU\text{-}ATT} = 0.07$, $\beta_{PU\text{-}ATT} = 0.24$). Besides methodological differences, the varying focus on certain samples and technologies may have caused these inconsistent findings and makes the findings less informative for education in general and teachers specifically. Hence, the types of samples and the specificity of technology are considered powerful moderators of TAM effects (see Table 2).

The current meta-analysis synthesizes the existing body of empirical research on the TAM for pre- and in-service teachers. It exploits the potential that lies within multivariate meta-analysis and synthesizes correlation matrices with the help of correlation-based MASEM—a MASEM approach that accounts for the dependencies between correlations within correlation matrices (M. W.-L. Cheung, 2015). We believe that this meta-analysis will stimulate the application of MASEM in educational research. Four interrelated research questions are addressed:

1. To what extent does an overall correlation matrix representing the relations among the TAM constructs show significant variation across studies? *(Fixed-versus random-effects models)*

2. To what extent does the TAM fit the data? Which of the hypothesized relations in the TAM can be established empirically based on the pooled correlation matrix? *(Structural equation models with and without direct effects; Models 1 and 2)*

3. To what extent do sample origin, teacher experience, and the specificity of technology affect the overall fit and the relations exhibited in the TAM? *(Subgroup analyses; Models 1 and 2)*

4. To what extent do external variables, including subjective norms, computer self-efficacy, and facilitating conditions explain variation in perceived usefulness and perceived ease of use? (*External variables; Models 3 and 4*)

Overall, our study follows the core steps of meta-analyses as it synthesizes the measures of associations between the TAM variables and quantifies their variation between studies first (Research Question 1), tests specific assumptions on the structural part of the TAM (Research Question 2), explores possible moderation of these assumptions by considering subgroups of teacher samples (Research Question 3), and finally tests the effects of alternative variables on the key TAM variables (Research Question 4).

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Section snippets

Literature search

A search in the following databases was conducted to identify the literature relevant to this meta-analysis: ERIC (Educational Resources Information Center), Learn Tech Lib

(Learning & Technology Library), PsycINFO, ScienceDirect, ProQuest Dissertation and Theses Database, IEEE Xplore Digital Library, ACM Digital Library, and Google Scholar (first 100 entries as of March 17, 2017). We used the following search terms and Boolean operators for ERIC and PsycINFO: ("Technology acceptance model" OR …

Description of studies

Table 3 presents the discrete characteristics of teacher samples, study methods, and characteristics for the $n = 114$ eligible TAM studies which provided $k = 124$ samples (i.e., 124 correlation matrices). For a more detailed presentation of these characteristics per study, we kindly refer the reader to the Supplementary Material S2. Overall, the samples described in these studies included pre- and in-service teachers, almost to the same extent. Moreover, the educational level teachers were …

Model fit and relations within the TAM

Overall, our meta-analysis of the relations within the TAM has shown that considerable variation in correlation matrices across study samples exists. This finding has at least two consequences: First, synthesizing correlation matrices should be based on random-effects models rather than fixed-effects models—this conclusion has been drawn in other domains as well (M. W.-L. Cheung & Cheung, 2016). Second, it implies heterogeneity in TAM relations which can potentially be explained by further …

Conclusions

The current meta-analysis synthesized the existing body of research on pre- and in-service teachers' technology adoption based on the Technology Acceptance Model using random-effects, correlation-based MASEM under M. W.-L. Cheung and Chan's (2005) two-step modeling approach. This study has two main contributions: First, from a substantive perspective, the meta-analytic findings support the applicability of the TAM to teacher samples and clarify some inconsistencies of certain relations within …

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these reviews have greatly improved the clarity of presentation and reasoning. …

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